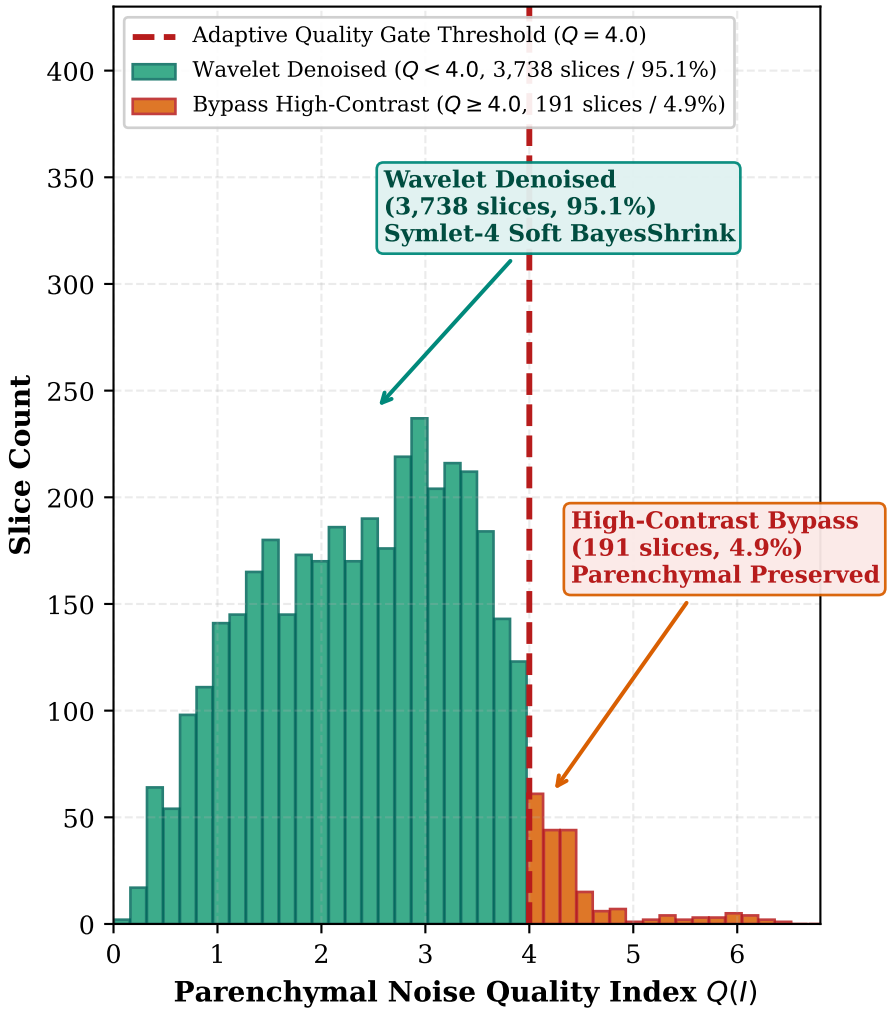
**(a) Quality Index $Q(I)$ Distribution & Adaptive Denoising Gate**



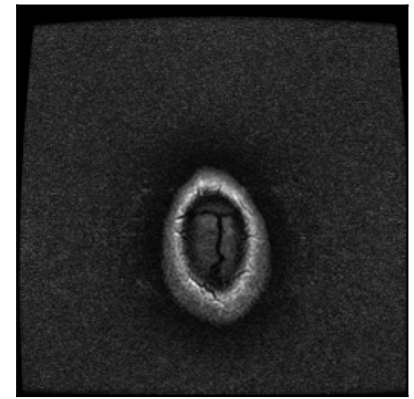

**(b) Excluded Artifact Slices (Top) & Wavelet Denoising Gate (Bottom)**

**Artifact 1: FOV Truncation**



$ED = 0.0023$, $CR = 0.0020$
$d_c = 0.64$ | **Excluded (0 px)**

**Artifact 2: Scanner Noise Fringe**



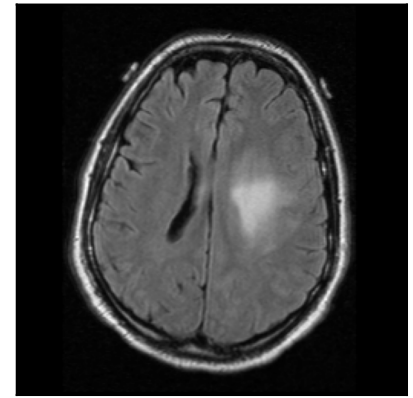
$ED = 0.0029$, $CR = 0.0024$
$d_c = 0.32$ | **Excluded (0 px)**

**Artifact 3: Non-Parenchymal Fringe**



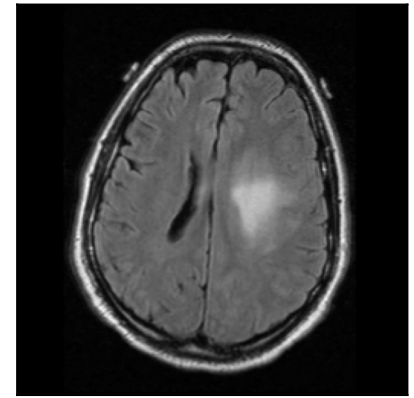
$ED = 0.0000$, $CR = 0.0000$
$d_c = 1.00$ | **Excluded (0 px)**

**Raw Slice (Moderate Noise)**



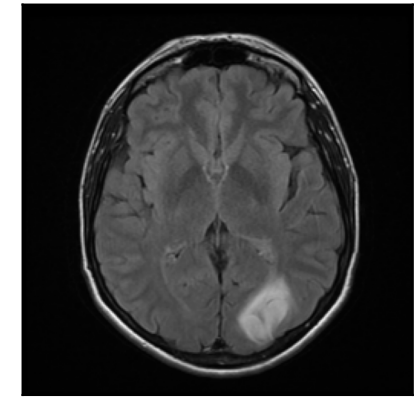
$Q(I) = 2.37 < 4.0$
**Raw Input Slice**

**Wavelet Denoised (Symlet-4)**



**BayesShrink Denoised**
**Artifacts Suppressed**

**High-Contrast Bypass**



$Q(I) = 4.12 \geq 4.0$
**Preserved (No Denoising)**